

In the renormalization group method you take a structure you don't understand and convert it to another structure you don't understand. You keep doing it until you finally understand.

These notes introduce the renormalization group (RG) through simple examples. We start with a cutoff problem in one-dimensional quantum mechanics, then turn to statistical mechanics: Kadanoff's block-spin argument and the scaling relations, Wilson's formulation in terms of recursion relations and fixed points, the exact decimation of the one-dimensional Ising chain, and the idea behind momentum-shell RG and the ϵ expansion. They are based on a 2020 project report at IIT Kanpur. Standard references are [Goldenfeld \(1992\)](#), [Cardy \(1996\)](#), [Chaikin and Lubensky \(1995\)](#) and [Pathria and Beale \(2011\)](#).

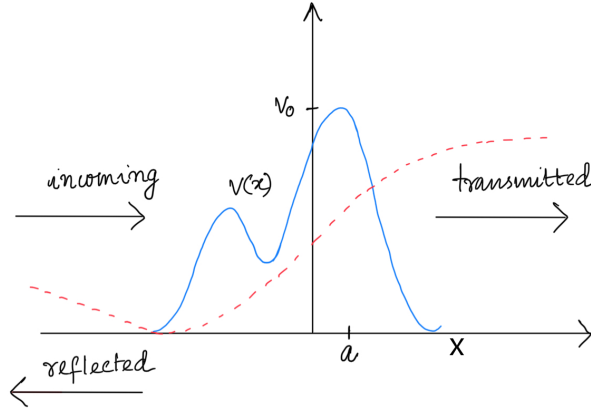
1 RG in quantum mechanics

Ultraviolet divergences, and the need for renormalization, appear already in simple quantum-mechanical problems ([Griffiths and Schroeter, 2018](#)). Consider a particle in one dimension with Hamiltonian

$$\hat{H} = \frac{1}{2}\hat{p}^2 + \hat{V}, \quad \hbar = m = 1, \quad (1.1)$$

where the potential has range $\sim a$ and height $\sim V_0$. We study scattering states,

$$\psi(x) = A e^{ipx} + B e^{-ipx} \quad (x \rightarrow -\infty), \quad \psi(x) = C e^{ipx} \quad (x \rightarrow +\infty). \quad (1.2)$$



Scattering on a short-range potential: the wavefunction varies on the scale $\lambda = 2\pi/p$ outside, and rapidly on the scale a inside.

Effective description. If the wavelength is long, $\lambda = 2\pi/p \gg a$, the details of the potential do not matter. For trial wavefunctions ϕ, ψ that vary only on scales much larger than a , expand $f(x) \equiv \phi^*(x)\psi(x)$ around $x = 0$:

$$\langle \phi | \hat{V} | \psi \rangle = \int dx V(x) f(x) = f(0) \int dx V(x) + f'(0) \int dx x V(x) + O(a^2/\lambda^2). \quad (1.3)$$

This is reproduced by the effective potential

$$V(x) \simeq c_0 \delta(x) + c_1 \delta'(x), \quad c_0 = \int dx V(x), \quad c_1 = - \int dx x V(x), \quad (1.4)$$

since $\int dx \delta'(x) f(x) = -f'(0)$. The constants c_0, c_1 are phenomenological couplings chosen to reproduce the true theory at low energy.

Leading term. For $c_0 \neq 0$ we keep only the delta function. The Schrödinger equation

$$-\frac{1}{2}\psi''(x) + c_0 \delta(x) \psi(x) = E \psi(x) \quad (1.5)$$

gives the jump condition $\psi'(0^+) - \psi'(0^-) = 2c_0\psi(0)$, and the transmission and reflection amplitudes

$$T = \frac{p}{p + ic_0}, \quad R = -\frac{ic_0}{p + ic_0}. \quad (1.6)$$

This is accurate for any short-range potential at low energy; for $p \rightarrow 0$, $T \simeq -ip/c_0$.

The problem. Now let the potential be odd, $V(-x) = -V(x)$. Then $c_0 = 0$ and the first nonvanishing term is $c_1\delta'(x)$. This term makes ψ itself discontinuous at the origin, and the Schrödinger equation with a δ' potential is not well defined as it stands. This inconsistency is the quantum-mechanical analogue of an ultraviolet divergence in field theory. Its origin is visible in the true solution: outside the potential the wavefunction varies on the scale λ , but inside it varies on the scale a , i.e. it contains high-momentum components that the effective description has thrown away.

Regularization. We restore a short-distance cutoff by replacing δ' with a finite difference,

$$V(x) = \frac{c_1}{2a} [\delta(x+a) - \delta(x-a)] \xrightarrow{a \rightarrow 0} c_1 \delta'(x). \quad (1.7)$$

For two delta functions of strengths $g_1 = g$ at $x = -a$ and $g_2 = -g$ at $x = +a$, with $g = c_1/2a$, matching the jump conditions gives

$$\frac{1}{T} = \left(1 + \frac{ig_1}{p}\right) \left(1 + \frac{ig_2}{p}\right) + \frac{g_1 g_2}{p^2} e^{4iap} = 1 + \frac{c_1^2}{4a^2 p^2} (1 - e^{4iap}). \quad (1.8)$$

At low energy, $pa \ll 1$, we have $1 - e^{4iap} \simeq -4iap$ and

$$\frac{1}{T} \simeq 1 - \frac{ic_1^2}{ap} \implies T \simeq \frac{p}{p + ic_R}, \quad \boxed{c_R = -\frac{c_1^2}{a}}. \quad (1.9)$$

The low-energy physics is that of Eq. (1.6) with an effective delta-function coupling c_R , which depends on c_1 and a only through this combination. Two features are worth noting:

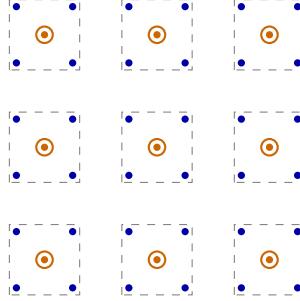
- The effective coupling is attractive for either sign of c_1 : an odd potential first acts at second order, and second-order corrections lower the energy.
- c_1 is dimensionless, while c_R has dimension $1/L$, like c_0 . The dimension is supplied by the cutoff a . A finite limit $a \rightarrow 0$ at fixed low-energy physics (c_R fixed) requires the bare coupling to scale as $c_1 \propto \sqrt{a}$. This is renormalization: physical predictions depend on a renormalized coupling, and the bare coupling must be adjusted as the cutoff is changed.

2 RG in statistical mechanics

Mean-field theory does not predict the critical exponents correctly, and attempts to improve it by perturbation theory fail because the corrections diverge as $T \rightarrow T_c$. Dimensional analysis then requires anomalous dimensions, which can only appear through a microscopic length (the lattice constant a), even though the correlation length is much larger. The scaling hypothesis follows from

the divergence of the correlation length, and the key insight is due to Kadanoff ([Kadanoff, 1966](#)): there is a relation between the couplings of an effective Hamiltonian and the length scale on which the order parameter is defined. Close to a critical point the correlation length ξ diverges, and the system becomes insensitive to a change of length scale; at the critical point it is completely insensitive. Wilson ([Wilson, 1971](#); [Wilson and Fisher, 1972](#)) completed this argument by showing how the couplings at different length scales can be computed explicitly. The RG also explains universality.

Kadanoff's block spins and scaling



Block spins for $l = 2$, $d = 2$: each block of $l^d = 4$ original spins (blue) is replaced by one block spin (orange) on a lattice with spacing la .

Consider N Ising spins $\sigma_i = \pm 1$ and group them into blocks of l^d spins ($l = 2$, $d = 2$ in the figure). Each block is represented by a block spin $\sigma' = \pm 1$, for example by a majority rule. There are $N' = l^{-d}N$ block spins on a lattice with spacing $a' = la$. To keep the density of degrees of freedom fixed, all lengths are rescaled as $r' = r/l$. Starting from

$$Z = \sum_{\{\sigma_i\}} e^{-\beta \mathcal{H}_N[\sigma]}, \quad (2.1)$$

we sum over the $N - N'$ internal degrees of freedom, leaving $Z = \sum_{\{\sigma'\}} e^{-\beta \mathcal{H}'_{N'}[\sigma']}$. If the new Hamiltonian has the same form with new reduced temperature t' and field h' , the singular part of the free energy satisfies $N' f_s(t', h') = N f_s(t, h)$, i.e.

$$f_s(t, h) = l^{-d} f_s(t', h'). \quad (2.2)$$

Close to the critical point t, h, t', h' are small, and the relation between them is linear,

$$t' = l^{y_t} t, \quad h' = l^{y_h} h, \quad (2.3)$$

so that

$$f_s(t, h) = l^{-d} f_s(l^{y_t} t, l^{y_h} h). \quad (2.4)$$

Since l is arbitrary, choose $l^{y_t} |t| = 1$:

$$f_s(t, h) = |t|^{d/y_t} f_s(\pm 1, h |t|^{-y_h/y_t}). \quad (2.5)$$

This is the scaling form of the free energy. Taking derivatives gives the critical exponents in terms of the two numbers y_t, y_h :

$$\alpha = 2 - \frac{d}{y_t}, \quad \beta = \frac{d - y_h}{y_t}, \quad \gamma = \frac{2y_h - d}{y_t}, \quad \delta = \frac{y_h}{d - y_h}, \quad \Delta = \frac{y_h}{y_t}, \quad (2.6)$$

with Δ the gap exponent. Under the same rescaling $\xi(t) = l\xi(l^{y_t}t)$, so $\xi \sim |t|^{-\nu}$ with $\nu = 1/y_t$, and the correlation function $G(r) \sim r^{-(d-2+\eta)}$ gives $\eta = d + 2 - 2y_h$. This yields the scaling relations

$$\alpha + 2\beta + \gamma = 2, \quad \gamma = \beta(\delta - 1), \quad d\nu = 2 - \alpha, \quad \gamma = \nu(2 - \eta). \quad (2.7)$$

Wilson's formulation

Kadanoff's argument motivates the scaling form but does not compute y_t , y_h or the scaling functions. It also assumes that coarse-graining preserves the form of the Hamiltonian, which is in general false: a nearest-neighbour model generates further-neighbour and multi-spin couplings after one step. Wilson addressed how all couplings change under repeated elimination of short length scales. Write the reduced Hamiltonian as

$$-\beta\mathcal{H} = \sum_n K_n \theta_n(S), \quad (2.8)$$

where $\theta_n(S)$ are local operators (functions of the degrees of freedom S) and K_n the corresponding couplings. After coarse-graining by a factor l the couplings change as

$$[K'] = R_l[K], \quad l > 1, \quad (2.9)$$

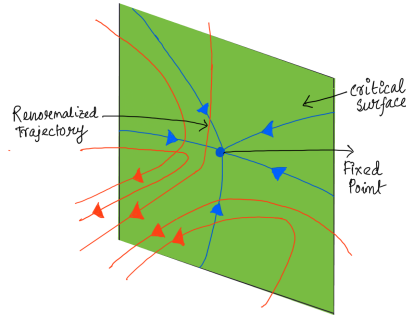
where R_l is the (in general nonlinear) recursion relation. Repeated application gives

$$K^{(n)} = R_l(K^{(n-1)}) = R_l^n(K^{(0)}), \quad \xi^{(n)} = l^{-n}\xi^{(0)}, \quad f_s^{(n)} = l^{nd}f_s^{(0)}. \quad (2.10)$$

Fixed points. A fixed point satisfies

$$R_l(K^*) = K^*, \quad \xi(K^*) = l^{-1}\xi(K^*), \quad (2.11)$$

so $\xi(K^*)$ is either 0 (a trivial, e.g. high- or zero-temperature fixed point) or ∞ (a critical fixed point). All points that flow into a critical fixed point have infinite ξ , since ξ only decreases along the flow; they form the critical surface. Points off the surface may first approach the fixed point but eventually flow away from it.



RG flow near a critical fixed point. Flows within the critical surface (green) end at the fixed point; flows off it are driven away.

Linearization and scaling fields. Near the fixed point write $K = K^* + k$ and linearize:

$$k' = M_l k, \quad (M_l)_{mn} = \left. \frac{\partial K'_m}{\partial K_n} \right|_{K^*}. \quad (2.12)$$

Let n_i be the eigenvectors of M_l with eigenvalues λ_i . Expanding $k = \sum_i u_i n_i$, the coefficients u_i (the scaling fields) transform independently,

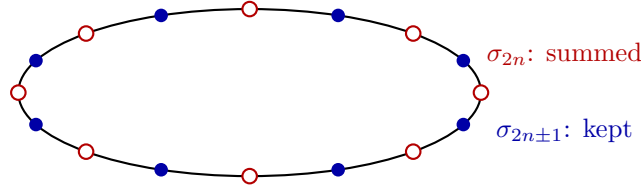
$$u_i^{(n)} = \lambda_i^n u_i^{(0)}, \quad \lambda_i = l^{y_i}, \quad (2.13)$$

where the second form follows from the group property $R_{l_1} R_{l_2} = R_{l_1 l_2}$. Three cases arise:

- (a) $\lambda_i > 1$ ($y_i > 0$): u_i grows under the flow and drives the system away from the fixed point. It is a *relevant* variable. A standard critical point has two, the reduced temperature $t = (T - T_c)/T_c$ and the reduced field $h = \mu B/k_B T$: $u_1 = a t + O(t^2)$, $u_2 = b h + O(h^2)$, with $y_1 = y_t$ and $y_2 = y_h$.
- (b) $\lambda_i < 1$ ($y_i < 0$): u_i decays and does not affect the critical behaviour. It is *irrelevant*, and gives corrections to scaling.
- (c) $\lambda_i = 1$ ($y_i = 0$): u_i is *marginal*. Its fate is decided by the next order in the expansion; marginal variables typically produce logarithmic corrections, or a continuously varying exponent.

The exponents (2.6) then follow from y_t and y_h . The hyperscaling relation $d\nu = 2 - \alpha$ fails for $d > 4$, where the exponents take their mean-field values; this is due to a dangerously irrelevant variable, the quartic coupling of the next section.

3 Exact RG for the one-dimensional Ising chain



Ising ring with periodic boundary conditions. Decimation sums over the even sites and leaves a ring of $N/2$ spins.

The one-dimensional Ising chain can be renormalized exactly by decimation (Pathria and Beale, 2011). With periodic boundary conditions,

$$Z_N(K_1, K_2) = \sum_{\{\sigma\}} \prod_{i=1}^N \exp \left[K_0 + K_1 \sigma_i \sigma_{i+1} + \frac{1}{2} K_2 (\sigma_i + \sigma_{i+1}) \right], \quad K_1 = \beta J, \quad K_2 = \beta \mu B. \quad (3.1)$$

The constant K_0 (initially 0) is needed because decimation generates a spin-independent term. Group the factors in pairs around each even site and sum over σ_{2n} :

$$\begin{aligned} \sum_{\sigma_{2n}=\pm 1} e^{2K_0 + K_1 \sigma_{2n} (\sigma_{2n-1} + \sigma_{2n+1}) + \frac{1}{2} K_2 (\sigma_{2n-1} + \sigma_{2n+1} + 2\sigma_{2n})} \\ = 2e^{2K_0} \cosh [K_1 (\sigma_{2n-1} + \sigma_{2n+1}) + K_2] e^{\frac{1}{2} K_2 (\sigma_{2n-1} + \sigma_{2n+1})}. \end{aligned} \quad (3.2)$$

Writing $\sigma'_j = \sigma_{2j-1}$, this must equal a factor of the original form,

$$e^{K'_0 + K'_1 \sigma'_j \sigma'_{j+1} + \frac{1}{2} K'_2 (\sigma'_j + \sigma'_{j+1})}, \quad (3.3)$$

for all four choices of $(\sigma'_j, \sigma'_{j+1})$. Solving the three independent conditions gives the recursion relations

$$\begin{aligned} e^{K'_0} &= 2e^{2K_0} \left[\cosh(2K_1 + K_2) \cosh(2K_1 - K_2) \cosh^2 K_2 \right]^{1/4}, \\ e^{K'_1} &= \left[\frac{\cosh(2K_1 + K_2) \cosh(2K_1 - K_2)}{\cosh^2 K_2} \right]^{1/4}, \\ e^{K'_2} &= e^{K_2} \left[\frac{\cosh(2K_1 + K_2)}{\cosh(2K_1 - K_2)} \right]^{1/2}. \end{aligned} \quad (3.4)$$

(These were checked by direct summation over the four spin configurations.) At $K_2 = 0$ the second relation reduces to the familiar $\tanh K'_1 = \tanh^2 K_1$. Since

$$Z_N(K_1, K_2) = e^{N' K'_0} Z_{N'}(K'_1, K'_2), \quad N' = N/2, \quad (3.5)$$

the free energy per spin, $f = -\ln Z_N/N$, obeys

$$f(K_1, K_2) = -\frac{1}{2} K'_0 + \frac{1}{2} f(K'_1, K'_2). \quad (3.6)$$

High-temperature fixed point. At $K_1 = K_2 = 0$ (infinite temperature) the couplings stay zero and $K'_0 = \ln 2$, so $f = -\frac{1}{2} \ln 2 + \frac{1}{2} f$, giving $f(0, 0) = -\ln 2$, the entropy of free spins, as it should be.

Fixed points. At $K_2 = 0$ the flow $\tanh K'_1 = \tanh^2 K_1$ has two fixed points: $K_1^* = 0$ with K_2 arbitrary (infinite temperature or $J = 0$, $\xi = 0$), and $K_1^* = \infty$, $K_2^* = 0$, reached at $B = 0$ and $T \rightarrow 0$, where $\xi = \infty$. All finite K_1 flow to $K_1 = 0$: there is no transition at $T > 0$.

Exponents at the zero-temperature fixed point. For large K_1 and small K_2 , Eq. (3.4) gives

$$K'_1 \simeq K_1 - \frac{1}{2} \ln 2, \quad K'_2 \simeq 2K_2. \quad (3.7)$$

Since $K_1^* = \infty$, introduce the variable $t = e^{-pK_1}$ ($p > 0$), which vanishes at the fixed point. Then

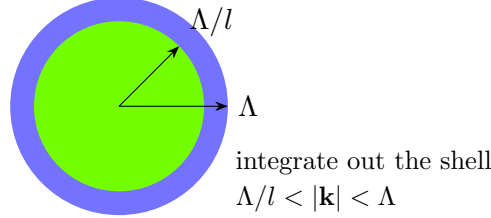
$$t' = 2^{p/2} t, \quad h' = 2h \quad (h \equiv K_2), \quad (3.8)$$

so with $l = 2$: $y_t = p/2$ and $y_h = 1$. With $d = 1$, Eq. (2.6) gives

$$\nu = \frac{2}{p}, \quad \Delta = \frac{2}{p}, \quad \alpha = 2 - \frac{2}{p}, \quad \beta = 0, \quad \delta = \infty, \quad \eta = 1. \quad (3.9)$$

The exponents depend on p because the choice of variable t is arbitrary at a zero-temperature fixed point. With $p = 2$, i.e. $t = e^{-2\beta J}$, the exact correlation length $\xi \simeq \frac{1}{2} e^{2\beta J}$ is reproduced as $\xi \sim t^{-1}$, $\nu = 1$. The values $\beta = 0$, $\delta = \infty$ and $\eta = 1$ reflect the step-function magnetization and the constant correlations of the ordered state at $T = 0$.

4 Momentum-space RG and the epsilon expansion



Momentum-shell RG. Instead of blocking spins in real space, we integrate out the Fourier components of the order parameter with wavenumbers in the shell $\Lambda/l < |\mathbf{k}| < \Lambda$, where $\Lambda \sim 1/a$ is the ultraviolet cutoff. Rescaling momenta by l and the field to restore the gradient term gives the recursion relations. Their Gaussian fixed point, $u_0^* = 0$, describes Landau theory. For $d < 4$ there is a second fixed point with $u_0^* \neq 0$, which in general is not small and cannot be reached by perturbation theory. Treating the dimension as a continuous variable, $d = 4 - \epsilon$, this fixed point has $u_0^* = O(\epsilon)$ and can be computed as a series in ϵ (Wilson and Fisher, 1972).

The idea behind the ϵ expansion. Start from the Landau–Ginzburg Hamiltonian for a scalar order parameter $m(\mathbf{r})$ (Ising universality class),

$$-\beta\mathcal{H}[m] = - \int d^d\mathbf{r} \left[\frac{1}{2}(\nabla m)^2 + \frac{1}{2}r_0 m^2 + \frac{1}{4}u_0 m^4 - h_0 m \right]. \quad (4.1)$$

The exponent is dimensionless. With $[\mathbf{r}] = L$, the gradient term gives $[m] = L^{(2-d)/2}$, and then

$$[r_0] = L^{-2}, \quad [u_0] = L^{d-4}, \quad [h_0] = L^{-(d+2)/2}. \quad (4.2)$$

Ignoring the feedback of other couplings, a rescaling by l therefore gives

$$u'_0 = l^{4-d}u_0 = l^\epsilon u_0, \quad \frac{du}{ds} = \epsilon u, \quad s = \ln l. \quad (4.3)$$

For $\epsilon > 0$ ($d < 4$) the quartic coupling grows: the Gaussian fixed point is unstable. For $d > 4$ it decays, Landau theory is correct, and u_0 is the dangerously irrelevant variable that spoils hyperscaling. The growth for $d < 4$ is cut off by the next term, generated by integrating out the shell,

$$\frac{du}{ds} = \epsilon u - A u^2 + O(u^3), \quad (4.4)$$

with a constant A . If $A < 0$, the only fixed point with $u \geq 0$ is $u^* = 0$. If $A > 0$ (as one-loop perturbation theory gives), there is a second fixed point,

$$\boxed{u^* = \frac{\epsilon}{A}}, \quad (4.5)$$

the Wilson–Fisher fixed point, which is stable along u and is $O(\epsilon)$, so perturbation theory around it is controlled. A fixed point with $u^* < 0$ is unphysical: the Landau free energy would be unbounded from below, since it could be lowered without limit by taking $|m| \rightarrow \infty$. At one loop the Wilson–Fisher fixed point gives, for the Ising class,

$$\nu = \frac{1}{2} + \frac{\epsilon}{12} + O(\epsilon^2), \quad \eta = O(\epsilon^2). \quad (4.6)$$

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